Hybrid Long-Term Multi Object Tracking (HLT-MOT) System

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***Abstract*—This paper presents a Hybrid Long-Term Object Tracking (HLT-MOT) system that effectively addresses the challenges of maintaining object identities over extended periods, particularly during occlusions or temporary disappearances. Our approach combines DeepSORT's frame-to-frame tracking capabilities with a robust re-identification framework powered by DINOv2 deep learning model. The system maintains feature galleries for tracked objects and employs both online and offline re-identification strategies to recover lost identities. Key innovations include: (1) a dual-model approach that balances accuracy and computational efficiency, (2) GPU-accelerated batch processing for feature extraction and matching, (3) Numba-optimized spatial-temporal consistency checks, and (4) dedicated mechanisms for prioritized tracking of a primary object. Experimental results demonstrate that our system significantly outperforms traditional tracking approaches in challenging scenarios involving long-term occlusions, crowded scenes, and appearance changes, while maintaining near real-time performance. The HLT-MOT system has important applications in surveillance, autonomous navigation, and human-computer interaction domains where persistent identity tracking is essential.**

***Index Terms*—Multi-Object Tracking, Long-Term Tracking, Re-identification, DeepSORT, Feature Extraction, Object Detection, Computer Vision**

I. Introduction

Robust, long-term person tracking is a cornerstone of intelligent systems in a multitude of applications, from human-robot collaboration and autonomous navigation to security and surveillance. The fundamental challenge lies not merely in following a person in an uncluttered view, but in maintaining a consistent identity through complex, real-world scenarios. These scenarios are characterized by frequent and prolonged occlusions, significant variations in appearance due to changing viewpoints and lighting, and the presence of visually similar distractors. Traditional tracking systems, which often couple a detector like YOLO [1] with a motion-centric tracker such as DeepSORT [2], [3], perform well in short-term scenarios but often fail when an object reappears after a long absence, as their underlying feature representations lack the robustness to bridge significant appearance gaps [4]. The recent emergence of large-scale, self-supervised foundation models has signified a new direction in visual perception, facilitating exceptional performance on open-set identification challenges. Works such as "Follow Anything" [5] have demonstrated the power of models like DINO [6] and SAM [7] to track arbitrary objects specified by a user query in real-time. Inspired by this progress, our work specializes these powerful representations for the specific and demanding task of long-term person re-identification (Re-ID). In this paper, we present a hybrid, real-time person tracking and re-identification system designed for long-term identity persistence. Our system synergistically combines the motion-prediction efficiency of DeepSORT for short-term tracking with a powerful appearance-based re-identification module built upon the DINOv2 [8]–[10] vision transformer. The core of our approach lies in a mask-aware feature extraction pipeline. By leveraging a segmentation-capable detector (YOLOv8-Seg) [11], [12], our system extracts discriminative features exclusively from the pixels of the segmented person, rather than the entire bounding box. This technique significantly improves feature quality and robustness against background clutter, a common failure point for traditional Re-ID methods. To ensure real-time applicability, the entire system is heavily optimized, employing TensorRT-accelerated [13] detection, batched GPU computations for feature extraction, and Numba-accelerated CPU operations.
Our main contributions are: A hybrid tracking architecture that integrates the motion-based tracking of DeepSORT with a robust DINOv2-based appearance gallery for long-term re-identification. A mask-aware feature extraction pipeline that utilizes segmentation masks to generate superior, clutter-free appearance descriptors for person Re-ID. An end-to-end, real-time optimized system that demonstrates effective long-term tracking of a primary individual through significant occlusions and appearance changes.

II. Related Work

The challenge of robust, long-term multi-object tracking (MOT) has been approached from several angles, evolving from motion-based methods to sophisticated deep learning architectures. Our work builds upon these established principles by integrating recent advances in self-supervised learning to address the specific demands of long-term person re-identification (Re-ID).

A. Tracking-by-Detection and DeepSORT

The dominant paradigm in modern MOT is tracking-by-detection. Early methods like SORT (Simple Online and Realtime Tracking) [14] relied solely on motion prediction with a Kalman filter and association based on spatial overlap (IoU). While fast, they are susceptible to failures during occlusions. A significant advancement came with DeepSORT [2], [3], which extended SORT by incorporating an appearance-based metric. DeepSORT uses a separate deep neural network to extract appearance features from each detected object. The association cost is then a weighted sum of motion similarity (Mahalanobis distance) and appearance similarity (cosine distance). This hybrid approach makes the tracker more robust to short-term occlusions and, as shown in systems like the person-following robot by Zhang et al. [4], forms a potent baseline for many tracking tasks.

B. Challenges in Long-Term Re-identification

While DeepSORT's appearance model improves short-term tracking, its lightweight Re-ID network often lacks the robustness to re-associate an object after a prolonged absence, significant viewpoint changes, or severe illumination variations. This is a common failure point in real-world scenarios, leading to identity switches and fragmented tracks. As noted by Zhang et al. [4], a fixed feature extractor often fails to generalize across the diverse appearances encountered during a long interaction, a problem known as domain drift. This limitation has spurred research into more powerful Re-ID techniques, but state-of-the-art models are too computationally expensive for direct integration into a real-time tracking loop without some optimization techniques.

C. Foundation Models for Visual Representation

The recent advent of large-scale, self-supervised foundation models has revolutionized visual perception. Models like DINO [6] and its successor, DINOv2 [8], learn highly discriminative and semantically rich visual features without human-annotated labels. These features exhibit remarkable robustness to viewpoint and appearance changes, making them ideal candidates for challenging Re-ID tasks. Works like ”Follow Anything” (FAn) by Maalouf et al. [5] have shown the potential of these models for open-set tracking. The FAn system leverages foundation models like DINO and the Segment Anything Model (SAM) [7] to track arbitrary, user-specified objects in real-time, highlighting a technique towards leveraging general-purpose, powerful visual representations.

D. Mask-Aware Feature Extraction

A key challenge in appearance-based tracking is the presence of background clutter within an object's bounding box, which can corrupt the feature vector. To mitigate this, some approaches utilize mask-aware feature extraction. By employing segmentation-capable detectors like YOLOv8-Seg [11], [12], it is possible to obtain precise pixel-level masks for each person. This allows feature extractors to compute embeddings exclusively from the pixels corresponding to the person, effectively reducing background noise. This principle, also seen in the FAn system's use of SAM [5], significantly enhances the discriminative power of appearance features, which is crucial for distinguishing between individuals in crowded environments.

III. System Architecture

The HLT-MOT system is designed as a modular, multi-stage pipeline that processes video frames to achieve robust long-term object tracking. The architecture's core principle is to synergize a high-speed, short-term tracker with a computationally intensive but highly accurate long-term re-identification (Re-ID) module. This dual approach allows the system to maintain track continuity in real-time while effectively recovering identities after prolonged occlusions or scene re-entry. The overall data flow is illustrated in Figure 1.

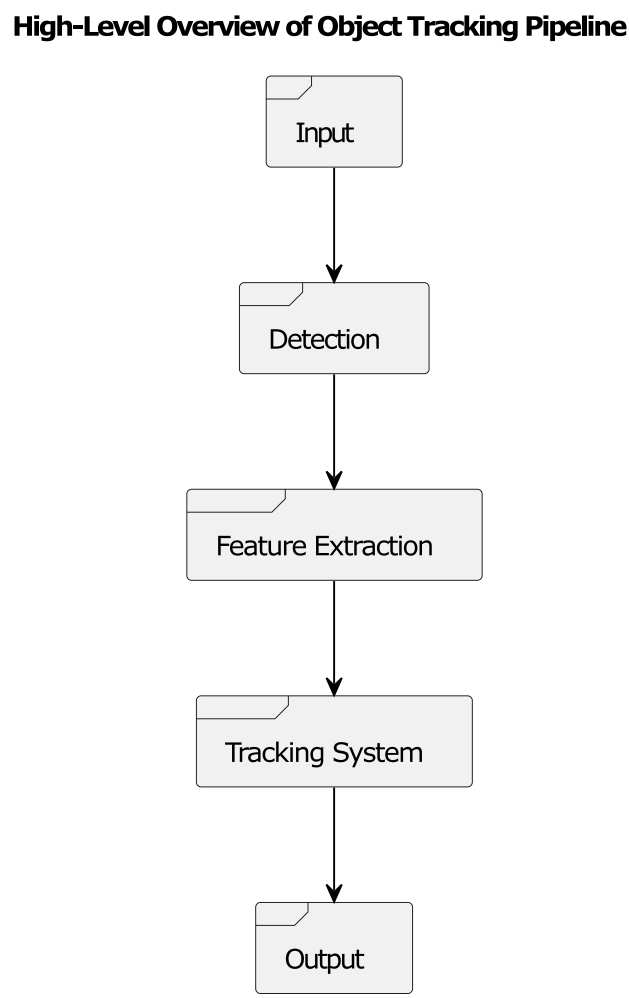

Fig. 1. High-level overview of the HLT-MOT system pipeline, from input frame to tracked output.

A. Architectural Pipeline

The tracking process is executed through a sequence of specialized components for each frame:

1) Detection and Segmentation: The pipeline begins with the YOLODetector, which processes the input frame to identify all persons and generate precise pixel-level segmentation masks for each one.

2) Appearance Feature Extraction: For each valid detection, the FeatureExtractor generates a discriminative appearance embedding. As shown in Figure, this process is mask-aware, using the segmentation mask to crop the person and isolate their features from background clutter, significantly improving the quality of the embedding.

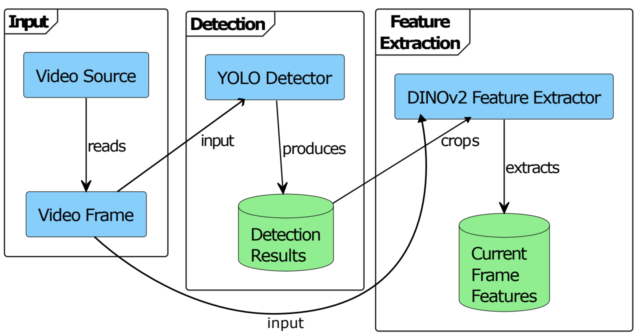

Fig. 2. Input processing pipeline: A frame is processed by the detector, and for each detected person, a mask-aware crop is sent to the feature extractor.

3) Short-Term Tracking: The detections, along with their spatial and appearance information, are passed to an integrated DeepSORT tracker. DeepSORT excels at frame-to-frame association using motion prediction and appearance cues, making it highly effective for maintaining identities through smooth trajectories and brief occlusions.

4) Long-Term Re-identification: The HybridTracker orchestrates the entire process and manages the crucial long-term Re-ID logic, as depicted in figure. It maintains a gallery of appearance embeddings for each object. When a track is lost and a new object appears, the HybridTracker compares the new object's features against the galleries of lost tracks. If a strong match is found, the object's original identity is recovered, bridging long temporal gaps caused by extended occlusions or the object leaving and re-entering the scene.

5) Primary Object Prioritization: The architecture includes a mechanism to give special priority to a designated "primary object" (typically the first one detected). This ensures that computational resources and Re-ID efforts are focused on maintaining the identity of this key target, a critical feature for applications like person-following.

6) Visualization: The final stage of the pipeline is visualizing the tracking results on the output video stream. For each processed frame, the system generates a display frame by drawing annotations directly onto a copy of the original image. To maintain clarity and focus, the visualization is centered on the designated primary object (ID 1) and avoids drawing other bounding boxes detected by Yolo. A green bounding box is drawn around this object, and its ID is displayed prominently above it. Additionally, key performance metrics and status information are overlaid on the frame, including the current frame number, the processing rate in frames per second (FPS), the status of the performance profiler, and whether the primary object is actively being tracked or is considered lost.

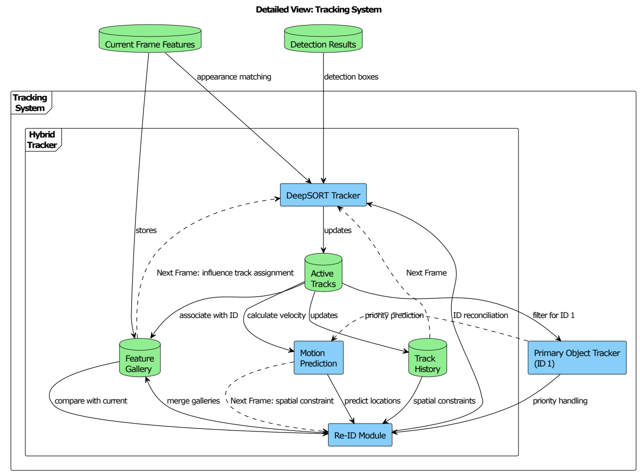

Fig. 3. The core tracking and Re-ID loop, managed by the HybridTracker.

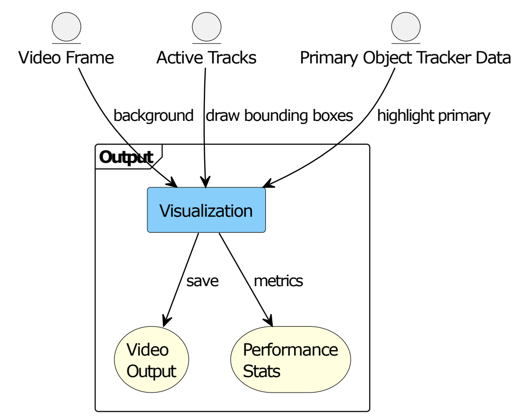

Fig. 4. Output visualization pipeline: The processed frame is annotated with primary ID and bounding box.

IV. Core Components

The HLT-MOT system is implemented as a collection of specialized Python classes, each responsible for a distinct part of the tracking pipeline. This modular design allows for clear separation of concerns and targeted optimizations. The following sections detail the core components and their implementation.

A. `YOLODetector`

This component implements the detection and segmentation stage of the pipeline. It serves as a streamlined wrapper around the `Ultralytics` library, specifically utilizing a `YOLOv8n-seg` model. The detector is configured to only identify objects of the "person" class (class ID 0 in the COCO dataset), reducing computational overhead. Its primary output is a list of detections, each containing bounding box coordinates, a confidence score, and a crucial pixel-level segmentation mask tensor. To maximize performance on compatible hardware, the class seamlessly integrates with NVIDIA's TensorRT, automatically loading pre-built .engine files for inference or generating them on-the-fly if they are not present.

B. `FeatureExtractor`

This module is responsible for generating the discriminative appearance embeddings used for Re-ID. It employs a pre-trained DINOv2 (ViT-B/14) model, loaded via torch.hub, which produces a robust 768-dimensional feature vector. Its core logic resides in the `extract_features_batch method`, which is heavily optimized for performance. The method takes a batch of detections (bounding boxes and their corresponding masks) and, for each one, applies the mask to the image crop to zero out background pixels. This ensures the resulting feature vector represents the person's appearance without environmental contamination. All masked crops are then stacked into a single tensor and passed through the DINOv2 model in one forward pass, fully leveraging GPU parallelism.

C. `DeepSORT Integration`

The system utilizes the `deep_sort_realtime` library to handle continuous, frame-to-frame tracking. DeepSORT's strength lies in its use of a Kalman filter to predict an object's position in the next frame, which it combines with an appearance cost to associate new detections with existing tracks. A key design choice in our system is to augment DeepSORT's native capabilities. While we provide the high-quality DINOv2 features to the algorithm, we do not rely on its internal, lightweight Re-ID model for long-term persistence. Instead, the more robust, gallery-based Re-ID is managed entirely by our HybridTracker, which uses the DINOv2 embeddings to handle challenges far beyond the scope of the standard DeepSORT algorithm.

D. `HybridTracker`

This class is the brain of the system, orchestrating the entire tracking and re-identification process. It encapsulates the core logic for ID management, track state maintenance, and the synergy between short-term and long-term tracking.

- **ID Management:** The tracker manages two layers of identification: the temporary track IDs generated by DeepSORT for frame-to-frame associations, and the system's own consistent, long-term IDs. An internal `id_mapping` dictionary bridges these two, ensuring an object's identity persists across disappearances.
- **Feature Gallery:** Its central data structure is the `feature_gallery`, which stores a `deque` (a fixed-size queue) of the most recent DINOv2 feature vectors for each long-term ID. This gallery serves as the memory of each object's appearance over time.
- **Re-identification Logic:** Re-ID is handled through a two-pronged approach. First, an *online* Re-ID check occurs whenever a new, untracked object appears. Its features are compared against the galleries of recently vanished tracks (`inactive_ids`) to see if it's a reappearance. Second, a more comprehensive *offline* Re-ID process (`_perform_offline_reid`) runs periodically. This function compares all active tracks with all inactive tracks using a batched GPU-based cosine distance calculation, allowing it to merge track fragments that belong to the same identity but were separated by a longer occlusion.
- **Spatial-Temporal Consistency:** Beyond appearance similarity, spatial and temporal cues are also considered. If the features are similar, it then enforces a spatial constraint. It checks if the new detection is physically close to the last known location of the lost track. This implementation choice effectively reduces false positives in Re-ID by using a search radius based on the last known position of the object and helps in crowded scenes where multiple objects may have similar appearances.
- **Primary Object Focus:** The tracker implements specialized logic to maintain the identity of a designated `primary_object_id`. This includes focused feature extraction for this specific target, making the system particularly suitable for applications like person-following where one object is of critical importance.

V. Optimization Techniques

Achieving real-time performance with a pipeline that includes a large vision transformer like DINOv2 requires aggressive optimization. Our system employs a multi-pronged strategy that leverages hardware acceleration, algorithmic efficiencies, and selective processing to minimize latency.

A. GPU Acceleration with TensorRT

The most computationally demanding components of the pipeline are offloaded to the GPU.

- **Accelerated Detection:** The YOLOv8-Seg detector is optimized using NVIDIA's TensorRT. The `YOLODetector` class is designed to automatically detect and load pre-built TensorRT `.engine` files for inference. If an engine file is not present, it is generated on-the-fly from the base PyTorch model, ensuring maximum inference speed on compatible hardware.
- **Feature Extraction:** The DINOv2 model for appearance feature extraction runs entirely on the GPU.
- **Re-ID Matching:** The cosine distance matrix calculation, used to compare features of new objects against the galleries of lost tracks during re-identification, is executed as a batched GPU operation, enabling rapid comparison of large feature sets.

B. Batch Processing

To fully exploit GPU parallelism, operations are batched wherever possible. Instead of processing detected objects sequentially, the `FeatureExtractor`'s `extract_features_batch` method collects all masked person crops from a single frame. These crops are stacked into a single tensor and passed through the DINOv2 model in one forward pass. This significantly reduces the overhead of individual model calls and is a key factor in maintaining high throughput. Similarly, the offline Re-ID process computes all necessary cosine distances in a single batched call.

C. Numba for CPU-Bound Operations

While the deep learning components are GPU-bound, certain geometric and logical operations are better suited for the CPU. To accelerate these, we use Numba, a Just-In-Time (JIT) compiler. Functions for calculating the Intersection over Union (IoU) between bounding boxes and the spatial distance between points are decorated with `@njit`. This compiles the Python code down to highly optimized machine code, providing a substantial speedup for these frequent calculations, which are critical for spatial consistency checks in the `HybridTracker`.

D. Selective and Prioritized Processing

Constant feature extraction for every detected object is computationally wasteful. The system incorporates several strategies to process data intelligently.

- **Periodic Re-identification:** The most expensive Re-ID logic, the *offline* check that compares all active tracks with all inactive ones, does not run on every frame. Instead, it is triggered at a configurable `re_id_interval`, striking a balance between identity recovery and computational load.
- **Primary Object Focus:** When a `primary_object_id` is designated, the tracker can enter an optimized mode. If the primary object is being tracked continuously, feature extraction is limited only to new detections that appear in close spatial proximity to the primary object's last known position. This drastically reduces unnecessary computations when the main target is in view.
- **Confidence-Based Filtering:** Detections below a certain confidence threshold are discarded early in the pipeline, preventing the system from wasting resources on low-quality or false-positive detections.

VI. Evaluation

To evaluate the performance of the HLT-MOT system, we tested the system on one of the standard multi-object tracking benchmark video called MOT16-06 [15], [16]. The evaluation focused on tracking performance under various conditions, including occlusions, crowded scenes, and appearance changes. We manually block the other individuals in the first frame to have a lock on the primary object before letting the original frames pass through the tracker. Since the system uses the first person it detects as the primary object, it could be automated as an interactive feature, like letting a user click to choose the target. A collage of a few intermediate video frames is shown in Figure 5, demonstrating how the system preserves the identity of the main object through reappearances and occlusions. Later in the video, as the lighting changes, the same person experiences another occlusion caused by several people passing in front of the main object.

Fig. 5. From top left and clockwise: (1) Women in white shirt and blue jeans tracked as primary, (2) Occluded by a person and tracker is lost, (3) Re-identified after reappearing

Fig. 6. From top left and clockwise: Same women in figure 5 tracked as primary walking and being occluded by a group of other people

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